ABOUT THE ORIGINAL STUDY

What educator applicants' short essays tell you about their potential job fit and performance

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Why It Matters

Short applicant essays may offer a low-cost way to learn about educator beliefs, priorities, and potential job fit. This study links essay ratings and machine learning–derived text features to hiring, retention, and student achievement, examining whether writing samples add useful information to educator screening.

Data and Sample

The study examines 10,384 applicant-by-year observations with essay information from one large urban California school district, covering March 2009–October 2015. Post-hire outcomes are observed only for hired educators, with value-added estimates available for about 450 observations in tested subjects and grades. Results may not generalize to other districts or today's AI-assisted applications.

Key Findings

- Essay quality and theme indicators identified by machine learning methods significantly predict essay scores, which in turn significantly predict teacher hiring status and on-the-job outcomes.
- A one-standard-deviation increase in HR-rated essay scores was associated with a 7.3-percentage-point increase in hiring probability and 1.9- and 2.7-percentage-point reductions in district departures within one and three years.
- Higher HR ratings were associated with higher student achievement, but estimates were small and statistical significance varied across specifications.
- Even after accounting for essay scores, writing quality, and other themes, five specific essay themes remain significantly correlated with the likelihood of being hired.
- Adding machine learning–derived features improved model explanatory power beyond HR ratings. Most themes were not significantly associated with retention or value-added; several suggestive achievement relationships did not survive multiple-testing adjustment.

Policy Implications

- Consider district-aligned short essays as one low-cost component of applicant screening. Review how successful applicants' themes relate to district priorities and applicant backgrounds to identify possible screening biases.
- Retain human review when using text-analysis tools. The authors emphasize HR evaluation and describe machine learning information as complementary.
- Address essay authorship before applying this approach today. The authors warn that AI-generated writing threatens screening based on applicant-generated text and discuss screening mechanisms or in-person writing, which may alter the applicant pool.

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